1. Research Foundation Abstract

Quantum Teleportation Scaling: From Laboratory to Reality

Teleport Massive Research Division | January 2026

1.1. Executive Summary

Seven peer-reviewed research papers demonstrate quantum teleportation has progressed from single-particle demonstrations to macroscopic-scale implementations. Research shows quantum teleportation works over 64-meter distances between chips (78.3% fidelity), 30.2-kilometer fiber links with classical signals (72.3% fidelity), and thermal microwave networks up to 4K (59.9% fidelity). All fidelities exceed classical thresholds, proving scalability from particles to chips, meters to kilometers, and ideal to real-world conditions.

1.2. Critical Breakthroughs

1. Chip-to-Chip (64m)

Fidelity: 78.3%

Loss: 0.32 dB/km

Proves teleportation works over macroscopic distances between separate quantum processors.

2. Fiber Network (30.2 km)

Fidelity: 72.3%

Coexistence: 400 Gbps

Demonstrates quantum and classical signals can share infrastructure.

3. Thermal Resilience (4K)

Fidelity: 59.9%

Channel: Thermal

Shows teleportation works in noisy, real-world environments.

4. Fundamental Physics

Connection: Black holes

System: Condensed matter

Reveals deep connections between teleportation and fundamental physics.

1.3. Scaling Pathway: From Particles to People

The Scaling Trajectory

Particles	Atoms	Chips	Devices	Macro
✓ Proven	Research	✓ 2026	Future	Vision
< 1 nm	1 nm	1 cm	10 cm	> 1 m

Current research demonstrates we are at the “Chips” stage (64m teleportation between superconducting chips). The pathway to macroscopic objects requires systematic scaling of entanglement resources.

1.4. Performance & Infrastructure Analysis

Fidelity vs. Distance

- 64 meters: 78.3%
- 30.2 kilometers: 72.3%
- Thermal channels: 59.9%

Key Insight:

Fidelity remains above classical threshold (50%) even at long distances and in noisy environments.

Infrastructure Compatibility

- Coexists with 400 Gbps classical
- Works in existing fiber networks
- Operates at 4K temperatures

Key Insight:

No need for dedicated quantum-only infrastructure. Can deploy in existing networks.

1.5. Technical Enablers & Strategic Conclusions

Critical Technologies

1. Ultralow-Loss Interconnects

0.32 dB/km loss enables long-distance teleportation.

2. Entanglement Scaling

Two-mode squeezed states and EPR pairs provide foundation for larger objects.

3. Noise Suppression

Wavelength optimization and filtering enable operation in noisy environments.

4. Error Correction

Current fidelities (70-80%) need improvement for safety-critical applications.

Implications

✓ **Feasibility Confirmed**

Quantum teleportation works at scales relevant for practical applications.

✓ **Infrastructure Ready**

Existing fiber networks can support quantum teleportation.

✓ **Systematic Approach Validated**

Incremental scaling (particles → atoms → chips → devices → macro) is the correct strategy.

The research foundation is solid.
Scaling quantum teleportation is not just possible—it's inevitable.

This abstract synthesizes findings from seven peer-reviewed research papers. Full papers follow.